

Problem 1)

a) Isolate the pulley:

$$2T = mg \Rightarrow T = \frac{mg}{2}$$

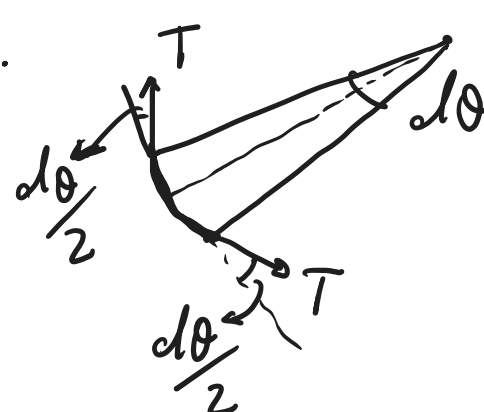
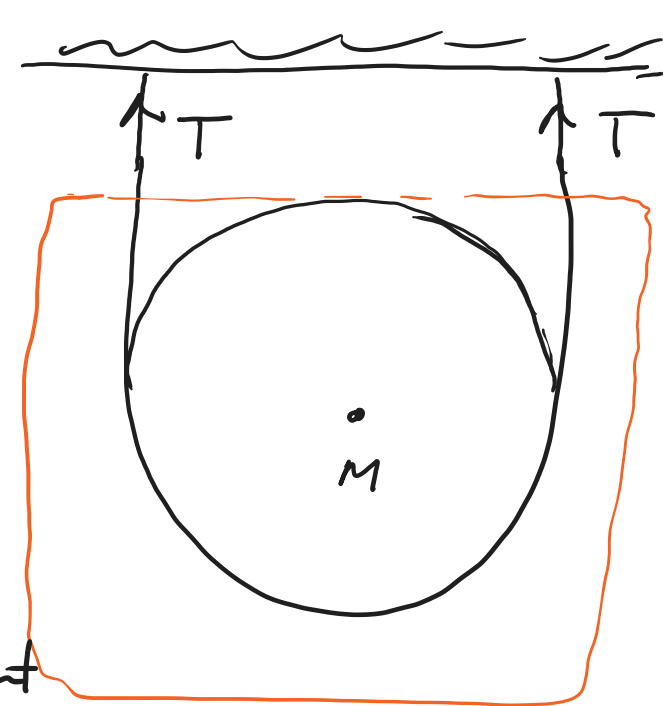
To obtain the normal force, consider a tiny element of the pulley subtending $d\theta$.

The tension is identical at both ends because string is massless and frictionless.

The normal force N is in response to the inward components of these two tangential tensions:

$$T_{\text{inward}} = T \sin \frac{d\theta}{2} \approx T \frac{d\theta}{2} \Rightarrow dN = N d\theta = 2T \frac{d\theta}{2}$$

$$\Rightarrow N = \frac{mg}{2} \Rightarrow \frac{N}{R} = \frac{mg}{2R}$$



(b) Because of friction, the tension is now differential. Thus, the lowest value of tension occurs at the lowest point of the pulley.

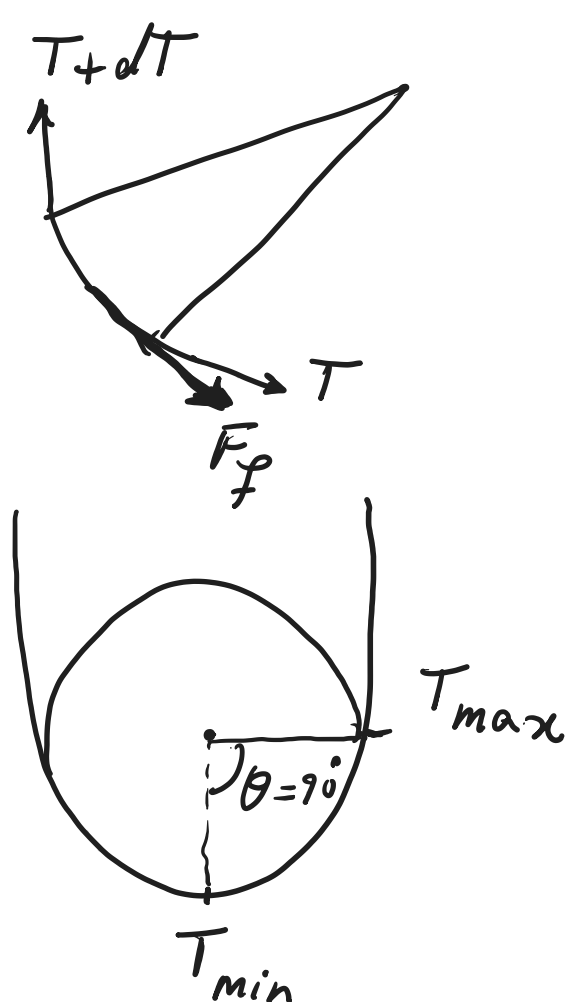
$$T + F_f = T + dT$$

$$F_f = \mu dN = \mu T d\theta \Rightarrow dT = \mu T d\theta$$

$$\Rightarrow \int_{T_{\min}}^{T_{\max}} \frac{dT}{T} = \mu \int_0^{\pi/2} d\theta \quad ; \quad T_{\max} = \frac{mg}{2}$$

$$\Rightarrow \ln T \Big|_{T_{\min}}^{T_{\max}} = \mu \frac{\pi}{2} \Rightarrow \frac{T_{\max}}{T_{\min}} = e^{\mu \pi/2}$$

$$\Rightarrow T_{\min} = \frac{mg}{2} e^{-\mu \pi/2}$$



Remark: $\mu \rightarrow 0 \Rightarrow T_{\min} \geq \frac{mg}{2}$ which makes sense from part (a).

$$\mu \rightarrow \infty \Rightarrow T_{\min} \geq 0$$

Problem 2)

known: v_0 init. velocity

unknown: prove

$$\vec{F} = -km\vec{v}$$

$$v = v_0 - kx$$

$$F = ma = m \frac{dv}{dt} = m v \frac{dv}{dx} = -kmv$$

$$\Rightarrow dv = -k dx \Rightarrow \int_{v_0}^{v(t)} dv = -k \int_0^{x(t)} dx$$

$$\Rightarrow v(t) = v_0 - kx(t)$$

Problem 3)

$$\vec{r} = x\hat{x} + y\hat{y} + z\hat{z}$$

$$\vec{B} = B_0\hat{y}$$

$$\vec{v} = \dot{x}\hat{x} + \dot{y}\hat{y} + \dot{z}\hat{z}$$

$$\vec{F} = q\vec{v} \times \vec{B} = m\vec{a}$$

$$\vec{a} = \ddot{x}\hat{x} + \ddot{y}\hat{y} + \ddot{z}\hat{z}$$

(x_0, y_0, z_0) is starting point.

$$\Rightarrow m(\ddot{x}\hat{x} + \ddot{y}\hat{y} + \ddot{z}\hat{z}) = q(\dot{x}\hat{x} + \dot{y}\hat{y} + \dot{z}\hat{z}) \times (B_0\hat{y})$$

$$= qB_0(\dot{x}\hat{z} - \dot{z}\hat{x})$$

$$\Rightarrow \begin{cases} m\ddot{x}\hat{x} = -qB_0\dot{z}\hat{x} \\ m\ddot{y}\hat{y} = 0 \\ m\ddot{z}\hat{z} = +qB_0\dot{x}\hat{z} \end{cases} \rightarrow \ddot{y} = 0 \Rightarrow \dot{y} = \dot{y}_0 = \text{const.} \Rightarrow y = \dot{y}_0 t + y_0$$

For simplicity: $\gamma = \frac{qB_0}{m}$

$$\begin{cases} \ddot{x} = -\gamma \dot{z} \Rightarrow \ddot{x} = -\gamma \dot{z} \\ \ddot{z} = \gamma \dot{x} \Rightarrow \ddot{z} = \gamma \dot{x} \end{cases} \rightarrow \text{Substitute the later eqns. in the former eqns.}$$

$$\Rightarrow \begin{cases} \ddot{\frac{x}{\gamma}} = -\dot{z} \\ -\ddot{\frac{z}{\gamma}} = \dot{x} \end{cases} \Rightarrow \begin{cases} \ddot{z} = -\gamma^2 \dot{z}, \text{ let } \omega = \dot{z} \\ \ddot{x} = -\gamma^2 \dot{x}, \text{ let } \eta = \dot{x} \end{cases} \Rightarrow \begin{cases} \ddot{\omega} = -\gamma^2 \omega \\ \ddot{\eta} = -\gamma^2 \eta \end{cases}$$

$$\Rightarrow \begin{cases} \omega = \alpha_\omega \sin(\gamma t) + \beta_\omega \cos(\gamma t) \\ \eta = \alpha_\eta \sin(\gamma t) + \beta_\eta \cos(\gamma t) \end{cases} \Rightarrow \begin{cases} z = \alpha_z \sin(\gamma t) + \beta_z \cos(\gamma t) + z_0 \\ x = \alpha_x \sin(\gamma t) + \beta_x \cos(\gamma t) + x_0 \end{cases}$$

But recall $\alpha_x, \alpha_z, \beta_x, \beta_z$ are connected through eqns above:

$$\begin{cases} \ddot{x} = -\gamma \dot{z} \\ \ddot{z} = \gamma \dot{x} \end{cases} \Rightarrow \begin{cases} \alpha_z = \beta_x = \alpha \\ \alpha_x = -\beta_z = \beta \end{cases} \Rightarrow \begin{cases} x - x_0 = \alpha \cos(\gamma t) + \beta \sin(\gamma t) \\ y - y_0 = \dot{y}_0 t \\ z - z_0 = -\beta \cos(\gamma t) + \alpha \sin(\gamma t) \end{cases}$$

The initial conditions, $t=0, \dot{z}(t=0) = \dot{z}_0; \dot{x}=0$ further yield,

$$\begin{cases} \gamma\beta = 0 \\ \gamma\alpha = \dot{z}_0 \end{cases} \Rightarrow \begin{cases} x - x_0 = \frac{\dot{z}_0}{\gamma} \cos(\gamma t) \\ y - y_0 = \dot{y}_0 t \\ z - z_0 = \frac{\dot{z}_0}{\gamma} \sin(\gamma t) \end{cases}$$